The Æther-flow Interpretation of Relativity

Substrate-Response / Self-Response Charge-Polarization Candidate
Exact-Preservation Observer-Equation Exact Absorbability Upgrade or Maximality Note

Abstract

The immediately preceding one-metric note proved the sharp exact observer-side theorem currently in hand on the fixed exact-preservation branch: the displayed burden

$$\mathfrak{B}^{\mathrm{obs}}_{\mu\nu} = \mathcal{D}^{\mathrm{res}}_{\mu\nu} + \mathcal{M}^{\mathrm{cmp}}_{\mu\nu} + \mathcal{Q}^{\mathrm{cmp}}_{\mu\nu} - 8\pi G_N\, \Delta T^{\mathrm{cmp}}_{\mu\nu}$$

does not survive as additional independent observer-side physics, so the completed branch is exactly observer-side effectively equivalent to Einstein–Hilbert plus ordinary matter through the same operative metric. That theorem still leaves one exact upgrade question open: can those same displayed sectors be absorbed exactly into the same Einstein–matter package, or does the branch presently stop at effective equivalence?

The present note attacks only that remaining question. It isolates two exact upgrade routes on the same one-metric branch. The first is direct absorbability: the already displayed burden must vanish in the exact observer quotient by the already-extracted gauge-exact and constraint-exact sectors. The second is an explicit same-package field-redefinition theorem: one must exhibit an exact invertible rewriting of the completed one-metric equation that removes the burden without introducing a second operative metric or a second matter law. The note proves that either route would upgrade the branch to strict same-package observer-side closure.

It then records the current outcome of that attack. The displayed sectors are indeed exact re-expansion terms of the same Einstein–matter package and remain exactly non-independent, but the presently recorded fixed package still does not contain either a direct absorbability identity or an explicit same-package field-redefinition theorem. Therefore strict observer-side closure is not yet proved on the current branch. The exact observer-side effective-equivalence theorem remains the strongest observer-side result currently proved on this line. If a later note still cannot supply either upgrade route, then the next honest move will be a genuine necessity or maximality theorem; the present note does not yet claim that stronger no-go.

1. Introduction

The previous note settled the sharpest exact theorem then available on the same one-metric charge-polarization branch. The displayed sectors inside $\mathfrak{B}^{\mathrm{obs}}_{\mu\nu}$ were shown not to survive as additional independent observer-side physics. That was a real gain. It removed the remaining burden sectors from the category of extra observer-side laws and left a much narrower open question:

> can those same sectors now be absorbed exactly into the same Einstein–matter package, or is the branch presently limited to exact observer-side effective equivalence?

This note addresses only that question. It does not introduce a deeper origin layer, a new completion solve, or a second operative metric. It keeps fixed the same candidate metric, the same

completed metric, the same completion solve, and the same ordinary matter route, and asks what exact upgrade statements those already recorded data do and do not yet support.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim that GR has now been derived from substrate microphysics. It does not claim the strict tensor identity $\mathfrak{B}^{\rm obs}_{\mu\nu} = 0$. Its positive role is narrower: it isolates the explicit exact upgrade routes that would close the remaining observer-side gap and states honestly that those routes are not yet supplied by the currently recorded fixed package.

2. Fixed One-Metric Package

Definition 1 (Fixed one-metric exact-preservation package)**.** *The present note keeps the following observer-level data fixed.*

1. *The candidate and completed operative metrics are*

$$g^{\rm cand} = g^{\rm br} + \Sigma^{\rm IER} + \Sigma^{\rm resp}, \qquad g^{\sharp} = g^{\rm br} + \Sigma^{\rm IER} + \Sigma^{\rm resp} + \Sigma^{\rm cmp}. \tag{1}$$

2. *The fixed completion solve is*

$$\mathcal{L}_{g^{\rm br}}(\Sigma^{\rm cmp}) = -\mathcal{V}^{\rm res}_{\mu\nu}. \tag{2}$$

3. *The completed observer equation is*

$$G_{\mu\nu}[g^{\sharp}] = 8\pi G_N\, T^{\rm matter}_{\mu\nu}[g^{\sharp}, \psi] + \mathcal{G}_{g^{\rm br}}(\Sigma^{\rm cmp})_{\mu\nu} + \mathfrak{B}^{\rm obs}_{\mu\nu}, \tag{3}$$

   with

$$\mathfrak{B}^{\rm obs}_{\mu\nu} \equiv \mathcal{D}^{\rm res}_{\mu\nu} + \mathcal{M}^{\rm cmp}_{\mu\nu} + \mathcal{Q}^{\rm cmp}_{\mu\nu} - 8\pi G_N\, \Delta T^{\rm cmp}_{\mu\nu}. \tag{4}$$

4. *The ordinary matter route is still exactly the same one-metric route*

$$\Delta T^{\rm cmp}_{\mu\nu} \equiv T^{\rm matter}_{\mu\nu}[g^{\sharp}, \psi] - T^{\rm matter}_{\mu\nu}[g^{\rm cand}, \psi]. \tag{5}$$

5. *The inherited candidate-branch decomposition*

$$\mathcal{D}^{\rm res}_{\mu\nu} = \mathcal{M}^{\rm mix}_{\mu\nu} + \mathcal{Q}^{\rm resp}_{\mu\nu} - 8\pi G_N\, \Delta T^{\rm cand}_{\mu\nu} \tag{6}$$

   is kept fixed.

6. *Gauge bookkeeping is already extracted explicitly in* (3)*, and the constraint-exact elimination sectors have already been removed before* $\mathfrak{B}^{\rm obs}_{\mu\nu}$ *is defined.*

7. *The previously established observer-side theorem on the same branch is that* $\mathcal{D}^{\rm res}_{\mu\nu}$*,* $\mathcal{M}^{\rm cmp}_{\mu\nu}$*,* $\mathcal{Q}^{\rm cmp}_{\mu\nu}$*, and* $\Delta T^{\rm cmp}_{\mu\nu}$ *are exactly non-independent observer sectors and that the completed branch is exactly observer-side effectively equivalent to Einstein–Hilbert plus ordinary matter.*

Remark 1. *Definition 1 is not a new package. It is only the exact observer-level package already recorded by the preceding note, restated here so the remaining upgrade question can be attacked directly on the same one-metric branch.*

Definition 2 (Exact observer quotient)**.** *For the fixed package of Definition 1, the* exact observer quotient *is the quotient of observer-side tensors by the already-extracted null classes:*

1. *gauge-exact bookkeeping tensors;*

2. *constraint-exact elimination tensors.*

The observer-quotient class of a tensor $X_{\mu\nu}$ *is denoted* $[X_{\mu\nu}]_{\rm obs}$*.*

Remark 2. *Definition 2 uses only the null sectors already recognized on this branch before the present note begins. It does* not *add exact non-independence or an unspecified field-redefinition class as new quotient relations.*

3. Two Exact Upgrade Routes

Definition 3 (Direct exact absorbability)**.** *The burden* $\mathfrak{B}^{\rm obs}_{\mu\nu}$ *is* directly exactly absorbable *on the fixed package if*

$$\left[\mathfrak{B}^{\rm obs}_{\mu\nu}\right]_{\rm obs} = 0. \tag{7}$$

Equivalently, there exist already-extracted gauge-exact and constraint-exact tensors $\mathfrak{G}^{\rm ex}_{\mu\nu}$ *and* $\mathfrak{C}^{\rm ex}_{\mu\nu}$ *such that*

$$\mathfrak{B}^{\rm obs}_{\mu\nu} = \mathfrak{G}^{\rm ex}_{\mu\nu} + \mathfrak{C}^{\rm ex}_{\mu\nu}. \tag{8}$$

Proposition 1 (Direct absorbability upgrades the completed observer equation)**.** *If* $\mathfrak{B}^{\rm obs}_{\mu\nu}$ *is directly exactly absorbable in the sense of Definition 3, then the completed observer equation closes in the exact observer quotient onto the same Einstein–matter package carried by* $g^\sharp$*:*

$$\left[G_{\mu\nu}[g^\sharp] - 8\pi G_N\, T^{\rm matter}_{\mu\nu}[g^\sharp, \psi]\right]_{\rm obs} = 0. \tag{9}$$

Proof. Insert (8) into (3). The terms beyond the Einstein–matter sector are then precisely gauge-exact and constraint-exact tensors, so they vanish in the quotient of Definition 2. This gives (9). □

Definition 4 (Explicit same-package field-redefinition upgrade)**.** *An* explicit same-package field-redefinition upgrade *on the fixed package means an explicit theorem exhibiting an exact invertible rewriting of the completed one-metric equation in terms of a rewritten metric variable* $\widehat{g}_{\mu\nu}$ *such that:*

1. $\widehat{g}_{\mu\nu}$ *is determined only by the already fixed branch data and introduces no second operative metric and no second matter law;*

2. *the rewritten observer equation is exactly Einstein–Hilbert plus ordinary matter through that same rewritten one-metric route, up to the already-extracted gauge-exact and constraint-exact sectors.*

Proposition 2 (An explicit same-package field-redefinition theorem upgrades closure)**.** *If an explicit same-package field-redefinition upgrade in the sense of Definition 4 is established on the fixed package, then the completed branch is exactly equivalent to Einstein–Hilbert plus ordinary matter through that rewritten one-metric description.*

Proof. This is immediate from the definition of the upgrade route: the field redefinition is assumed to be exact, invertible, and to preserve the one-metric Einstein–matter package up to the already-extracted null sectors. Therefore the rewritten observer equation is exactly Einsteinian in that precise same-package sense. □

Theorem 1 (Exact upgrade criterion on the fixed branch)**.** *On the fixed one-metric exact-preservation branch, strict observer-side closure beyond the already-proved effective-equivalence theorem will be established if either of the following exact upgrade routes is proved:*

1. *direct exact absorbability of* $\mathfrak{B}^{\mathrm{obs}}_{\mu\nu}$ *in the sense of Definition 3;*
2. *an explicit same-package field-redefinition upgrade in the sense of Definition 4.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. □

Remark 3. *Theorem 1 is deliberately conditional. It isolates the two exact upgrade routes that the branch now needs; it does not claim that one of them has already been proved.*

4. What the Fixed Package Already Gives

Proposition 3 (The displayed sectors are exact re-expansions of the same Einstein–matter package)**.** *On the fixed branch of Definition 1:*

1. $\mathcal{D}^{\mathrm{res}}_{\mu\nu}$ *is the inherited candidate-branch re-expansion sector recorded in* (6)*;*
2. $\mathcal{M}^{\mathrm{cmp}}_{\mu\nu}$ *and* $\mathcal{Q}^{\mathrm{cmp}}_{\mu\nu}$ *are completion-generated Einstein re-expansion sectors built from the same one-metric completion package;*
3. $\Delta T^{\mathrm{cmp}}_{\mu\nu}$ *is the same-metric matter re-expansion recorded in* (5)*.*

Consequently every displayed term in $\mathfrak{B}^{\mathrm{obs}}_{\mu\nu}$ *belongs to the same one-metric Einstein–matter package already fixed on the branch.*

Proof. Item (1) is exactly the inherited decomposition (6). Items (2) and (3) are already part of the fixed-package description recorded in Definition 1 and in the preceding exact non-independence / effective-equivalence note. Therefore all displayed sectors are exact re-expansion sectors of one common Einstein–matter package rather than extra independent observer-side laws. □

Corollary 1 (Remaining exact observer-side question)**.** *Because the gauge-exact and constraint-exact sectors are already extracted and because the displayed burden sectors are already known to be exactly non-independent re-expansion sectors of the same Einstein–matter package, the remaining exact observer-side question on this branch is now narrow:*

does one have a new exact identity of the direct absorbability type (8)*, or a new explicit same-package field-redefinition theorem of the type isolated in Definition 4?*

Proof. Definition 1 records the already-extracted null sectors. Proposition 3 records the common same-package origin of the displayed sectors. The preceding note records their exact non-independence. Once those facts are fixed, the only remaining exact upgrade burden is whether one of the two routes of Theorem 1 can now be proved. □

5. Current Recorded Status

Proposition 4 (Current upgrade status on the recorded branch)**.** *At current manuscript scope, the recorded fixed package does* not yet *contain:*

1. *a proved direct absorbability identity* $\left[\mathfrak{B}^{\mathrm{obs}}_{\mu\nu}\right]_{\mathrm{obs}} = 0$*;*
2. *a proved explicit same-package field-redefinition upgrade theorem.*

Therefore strict observer-side closure is not yet proved on the current branch.

Proof. Definition 1 records the presently invoked exact identities and observer-side theorems from the same one-metric branch. Those recorded results establish exact non-independence and observer-side effective equivalence, but they do not supply (7). Nor do they display an explicit invertible rewritten metric variable $\widehat{g}_{\mu\nu}$ together with a proved Einstein–matter reduction theorem of the type isolated in Definition 4. Therefore neither exact upgrade route of Theorem 1 is yet established, and strict observer-side closure remains unproved at current scope. □

Corollary 2 (Current strongest proved observer-side result)**.** *On the present one-metric branch, the exact observer-side effective-equivalence theorem of the preceding note remains the strongest observer-side result currently proved.*

Proof. The preceding note proves exact observer-side effective equivalence. Proposition 4 shows that the stronger upgrade routes isolated in Theorem 1 are not yet established on the recorded branch. Therefore the effective-equivalence theorem remains the strongest observer-side theorem currently proved. □

Remark 4. *Corollary 2 is a current-status statement, not a universal no-go. The present note does not claim that no future argument from deeper or sharper structure could supply one of the missing upgrade routes.*

6. Project-Level Verdict on the Next Move

Theorem 2 (Exact next-step verdict)**.** *For the one-metric charge-polarization branch at current scope, the next honest primary move is to prove one of the two exact upgrade routes isolated in Theorem 1. More precisely:*

1. *either prove a direct absorbability identity making* $\left[\mathfrak{B}^{\mathrm{obs}}_{\mu\nu}\right]_{\mathrm{obs}} = 0$*;*
2. *or prove an explicit same-package field-redefinition theorem eliminating the remaining burden.*

A deeper-origin manuscript can count as primary on this question only if it contributes one of those exact upgrade ingredients. Merely preserving the same completed observer equation is not by itself sufficient.

Proof. Theorem 1 identifies the exact upgrade routes. Proposition 4 shows that neither route is yet established. Therefore the next honest primary burden is to supply one of them. A deeper-origin note that leaves the same completed observer equation unchanged may still be useful, but it resolves the present bottleneck only if it delivers one of those missing exact identities or theorems. □

Corollary 3 (Where a genuine maximality or no-go theorem would belong)**.** *If later work still cannot establish either exact upgrade route on the fixed one-metric branch, then the next honest step will be a separate genuine necessity, maximality, or no-go theorem. The present note does not yet claim that stronger verdict.*

Proof. This follows from Theorem 2 together with the current-status nature of Proposition 4 and Corollary 2. The present note isolates the missing upgrade routes and records that they remain open; a later impossibility theorem would require additional argument beyond that current-status result. □

7. Conclusion

The exact observer-side gap left by the preceding effective-equivalence note is now sharply isolated. The displayed burden sectors are already known not to survive as additional independent observer-side physics, and they are already known to be exact re-expansion sectors of the same one-metric Einstein–matter package. The only exact question that remains on this branch is whether those sectors can now be removed by a new direct absorbability identity or by an explicit same-package field-redefinition theorem.

The present note proves that either of those routes would upgrade the branch to strict observer-side closure, but it also records honestly that neither route is yet established by the currently recorded fixed package. Therefore strict observer-side closure is still open. The strongest observer-side theorem currently proved on this line remains the exact effective-equivalence theorem of the preceding note.

This is enough to make the next project-aligned move explicit. The next primary manuscript should not drift into deeper origin work merely because deeper origin material exists. It should prove a direct absorbability identity, prove an explicit same-package field-redefinition theorem, or else, if that cannot be done in later work, motivate a separate genuine maximality or no-go theorem. That is the narrow exact burden still standing between observer-side effective equivalence and literal Einstein closure on the present one-metric branch.